approaches zero, there is a sudden increase in the molar conductance and the curve is almost parallel to Λ_m axis. This is due to the fact that the dissociation of the weak electrolyte increases with the increase in dilution (Ostwald dilution law). Λ_m^0 values for strong electrolytes can be obtained by extrapolating the straight line, as shown in figure (9.4). But the same procedure is not applicable for weak electrolytes, as the plot is not a linear one, Λ_m^0 values of the weak electrolytes can be determined using Kohlraush's law.

9.2.1 Debye - Huckel and Onsager equation

We have learnt that at infinite dilution, the interaction between the ions in the electrolyte solution is negligible. Except this condition, electrostatic interaction between the ions alters the properties of the solution from those expected from the free – ions value. The influence of ion-ion interactions on the conductivity of strong electrolytes was studied by Debye and Huckel. They considered that each ion is surrounded by an ionic atmosphere of opposite sign, and derived an expression relating the molar conductance of strong electrolytes with the concentration by assuming complete dissociation. Later, the equation was further developed by Onsager. For a uni – univalent electrolyte the Debye Huckel and Onsager equation is given below.

$$\Lambda_m = \Lambda_m^0 - \left(A + B\,\Lambda_m^0\right)\sqrt{C} \qquad .....(9.12)$$

Where A and B are the constants which depend only on the nature of the solvent and temperature. The expression for A and B are

$$A = \frac{82.4}{\sqrt{DT}\,\eta} \quad ; \quad B = \frac{8.20 \times 10^5}{\sqrt[3]{DT}}$$

Here, D is the dielectric constant of the medium, η the viscosity of the medium and T the temperature in Kelvin.

9.2.2 Kohlraush's law

The limiting molar conductance Λ_m^0 is the basis for kohlraush law. At infinite dilution, the limiting molar conductivity of an electrolyte is equal to the sum of the limiting molar conductivities of its constituent ions. i.e., the molar conductivity is due to the independent migration of cations in one direction and anions in the opposite direction.

For a uni – univalent electrolyte such as NaCl, the Kohlraush's law is expressed as

$$\left(\Lambda_m^0\right)_{NaCl} = \left(\lambda_m^0\right)_{Na^+} + \left(\lambda_m^0\right)_{Cl^-}$$

In general, according to Kohlraush's law, the molar conductivity at infinite dilution for a electrolyte represented by the formula $A_x\ B_y$, is given below.

$$\left(\Lambda_m^0\right)_{A_xB_y} = x\left(\lambda_m^0\right)_{A^{y+}} + y\left(\lambda_m^0\right)_{B^{x-}} \qquad .....(9.13)$$

Kohlraush arrived the above mentioned relationship based on the experimental observations such as the one as shown in the table. These result show that at infinite dilution each constituent ion of the electrolyte makes a definite contribution towards the molar conductance of the electrolyte irrespective of nature of other ion with which it is associated

XII U9 Electro Chemistry.indd 41 11-01-2021 14:43:57